

ARG-VU: Affordance Reasoning with Physics-Aware 3D Geometry for Visual Understanding in Robotic Surgery

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Abstract—A surgical digital twin must do more than reconstruct the scene: it must tell a robot what a tool can safely do to tissue that is deformable, compliant, and coupled with tool motion. We present ARG-VU, a lightweight, physics-aware affordance layer that sits between perception and simulation. From stereo endoscopic video we obtain temporally tracked tissue geometry, embed local deformation constraints with extended position-based dynamics (XPBD) at sparse representative geometry points (RGPs), and read out an anisotropic stiffness metric from constraint sensitivity. A tool motion in $SE(3)$ is then scored by a physics-aware compliance energy (PACS) that measures whether the displacement it induces is admissible under local tissue mechanics; a purely kinematic score (PAS) serves as baseline. On a StereoMIS gallbladder-retraction sequence, the XPBD-derived compliant direction anticipates subsequent tissue motion well above a velocity-persistence baseline, and PACS stays concentrated near the tool-tissue contact while PAS fluctuates. We state what this early, single-sequence evidence does and does not show, and outline how the layer can feed twin construction and action checking.

I. INTRODUCTION

Surgical digital twins promise that a robot can reason about the operative field before it acts: perception says *what and where* things are, simulation says *what would happen next*, and learning closes the loop [1]–[3]. Between the two sits a question that neither answers cheaply: *what can the tool do to this tissue right now?* This is a question about affordance [4], [5].

Gap. Surgical perception has advanced on tool tracking [6] and scene reconstruction [7]–[9], but these outputs describe geometry and appearance; they do not say which tool motions the tissue will mechanically admit. Affordance methods in robotics rely on visual cues such as contact points and grasp poses, mostly for rigid objects [10], [11]; deformable-object variants remain visually driven [12], [13]. Full XPBD-based tissue simulation [1], [14] can answer the question, but it needs a tissue mesh and calibrated material parameters that a live endoscopic view does not provide.

Challenges. (C1) Tissue is deformable, anisotropic and coupled with the tool, so the same tool motion is easy along one direction and hard along another. (C2) Observations are noisy (specularities, tool occlusion, heartbeat), while local mechanics is sensitive to jitter. (C3) Neither tissue parameters nor affordance labels are available intra-operatively, and the method must be light enough for real-time use.

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TABLE I
POSITIONING OF ARG-VU (QUALITATIVE).

	Tissue mechanics	No mesh or material params	Tool-motion reasoning
Reconstruction [7]–[9]	–	✓	–
Visual affordance [10]–[13]	–	✓	✓
XPBD simulation [1], [14]	✓	–	✓
ARG-VU	✓	✓	✓

Approach. ARG-VU turns tracked tissue points into local, constraint-based mechanical models and asks whether a given tool motion is admissible under them (Fig. 1). It is deliberately not a full simulator: it extracts only the local compliance structure needed to judge a tool motion. Our contributions are:

- a geometry-centric representation that couples tracked surface points with XPBD [15] constraints at sparse *representative geometry points* (RGPs);
- a constraint-manifold stiffness metric obtained from Jacobian-based sensitivity analysis;
- a physics-aware affordance formulation that maps end-effector motion in $SE(3)$ to local tissue response, with a kinematic baseline for comparison.

A longer preprint is available [16]; this abstract adds the twin-oriented framing and the task-conditioning discussion of Sec. V.

II. PROBLEM SETTING

Input and output. Given stereo endoscopic video, we have tissue points $\mathcal{P}_t = \{p_j(t)\}_{j=1}^N$ tracked in the camera frame and a tool pose $T_t = [R_t \ o_t] \in SE(3)$ per frame. The output is a per-point affordance score $A_{i,t}$ for the tool motion between t and $t+1$.

What “affordance” means here. A tool motion is afforded at a tissue point if the displacement it induces there is *admissible under the local deformation constraints*, i.e., it lies (to first order) in the tangent space of the local constraint manifold rather than against its constraint normals. Affordance is therefore a property of the (tool motion, tissue mechanics) pair, not of appearance. **Scope:** the score is high where tissue yields to the tool, which suits grasp-and-retract tasks. Tasks that exploit resistance (puncture, dissection under tension) need a task-conditioned score, which we discuss in Sec. V.

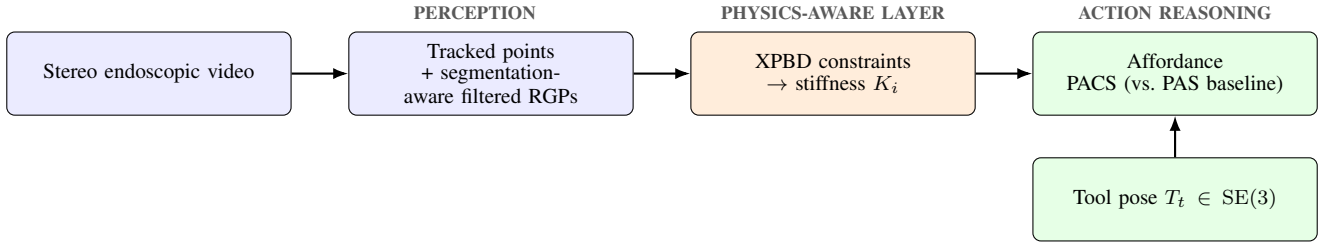


Fig. 1. ARG-VU overview. Tracked tissue geometry becomes local XPBD constraint models at RGPs; the resulting stiffness K_i scores tool-induced displacement. No tissue parameters or mesh are needed.

III. METHOD

A. Tracked Geometry and Representative Geometry Points

We obtain $\mathcal{P}_t$ from stereo disparity and the online reconstruction-and-tracking method of Hayoz *et al.* [17]. This backend is Gaussian-based [18] but is used only to obtain stable geometric observations; everything below depends on $\mathcal{P}_t$ alone and could use any tracker with persistent 3D points.

To keep computation small we operate on a sparse set of RGPs $\mathcal{P}_{\text{rep}} \subset \mathcal{P}_0$, selected in the first frame by radius-based clustering (radius r , minimum neighborhood size $N_{\min}$); RGP identities stay fixed over time. Each RGP i has a time-varying neighborhood

$$\mathcal{N}_t(i) = \{p_j(t) \in \mathcal{P}_t : \|p_j(t) - p_i(t)\| \leq r\}. \quad (1)$$

Neighborhoods overlap and points contribute through continuous kernel weights, so RGPs are soft control nodes rather than a hard partition, and no mesh is built. Because touching organs have very different compliance, we filter neighborhoods with per-frame 2D segmentation labels $S(\cdot, t)$ projected to 3D:

$$\mathcal{N}'_t(i) = \{p_j(t) \in \mathcal{N}_t(i) : S(p_j, t) = S(p_i, t)\}. \quad (2)$$

B. XPBD Constraints and Stiffness Metric

Each point gets a smoothed rest position $\bar{p}_j^0$ (temporal average over a short window). For RGP i the local deformation gradient is the weighted least-squares fit

$$F_i = \arg \min_F \sum_{p_j \in \mathcal{N}'_t(i)} w_{ij} \|(p_j(t) - p_i(t)) - F(\bar{p}_j^0 - \bar{p}_i^0)\|^2, \quad (3)$$

with $w_{ij} = \exp(-\|\bar{p}_j^0 - \bar{p}_i^0\|^2/\sigma^2)$. Elasticity is enforced through a hydrostatic constraint $C_{\text{hydro}} = \det F_i - 1$ (near volume preservation) and a deviatoric constraint $C_{\text{devia}} = \text{tr}(F_i^\top F_i) - 3$ (shear and distortion), solved with a quasi-static XPBD scheme [14], [15] using Gauss–Seidel updates and adaptive clamping (98th percentile) against outlier tracking errors.

Let $r_i = [C_{\text{hydro}}, C_{\text{devia}}]^\top$ and $J_i = \partial r_i / \partial p_i \in \mathbb{R}^{2 \times 3}$. Displacements in $\text{Null}(J_i)$ preserve the constraints to first order (the tangent space of the constraint manifold, i.e., compliant motions); displacements along $\text{Range}(J_i^\top)$ (constraint normals) are resisted. We encode this as

$$K_i = J_i^\top J_i + \varepsilon I_3. \quad (4)$$

Since J_i has two rows, K_i generically has a single compliant direction $e_i^{\min}$ (smallest eigenvalue) and two stiff ones.

C. Affordance from Tool Motion

With incremental twist $\xi_t = [v_t^\top, \omega_t^\top]^\top$, $v_t = o_{t+1} - o_t$ and $\omega_t = \text{Log}(R_{t+1}R_t^\top)^\vee$, the rigidly induced displacement at RGP i is

$$\Delta p_{i,t} = v_t + \omega_t \times (p_i(t) - o_t). \quad (5)$$

The physics-aware compliance energy and score (PACS) are

$$E_{i,t} = \frac{1}{2} \Delta p_{i,t}^\top K_{i,t} \Delta p_{i,t}, \quad A_{i,t} = -E_{i,t}. \quad (6)$$

Low energy means the induced motion follows compliant directions and is mechanically supported; high energy means it works against the local constraint structure. To suppress pose jitter and transient contact we smooth with an exponential moving average, $\bar{E}_{i,t} = \lambda \bar{E}_{i,t-1} + (1 - \lambda) E_{i,t}$ and $\bar{A}_{i,t} = -\bar{E}_{i,t}$.

Kinematic baseline (PAS). PAS ignores stiffness. Over a window τ , let $\Sigma_{i,t}^{\text{traj}}$ be the covariance of the tracked point's own displacements and $d_a = v_t / \|v_t\|$ the tool action direction; then $E_{i,t}^{\text{PAE}} = d_a^\top \Sigma_{i,t}^{\text{traj}} d_a$ and $A_{i,t}^{\text{PAS}} = -E_{i,t}^{\text{PAE}}$. PAS therefore reflects how tissue motion co-varies with the tool, without any notion of whether that motion is mechanically admissible.

IV. EXPERIMENTS

A. Data and Tool Pose

We use one gallbladder-retraction sequence from StereoMIS [19]: 200 frames at 30 fps, 640×512 , of which 125 involve active tool–tissue interaction. Depth comes from the tracker of [17]. For each frame we track K tool keypoints with LocoTrack [20], lift them to 3D with depth, and take the centroid as translation and the principal axis (PCA, sign fixed by temporal continuity) as orientation. This pose is coarse, and we treat it as a limitation.

B. Affordance Maps: PACS versus PAS

Fig. 2 shows a pulling action on the gallbladder. When the tool is still approaching (left), both maps are broad: many locations are compatible with the tool pose. As retraction begins and proceeds, PACS concentrates around the tool–tissue interaction region and stays there across frames, whereas PAS changes markedly between frames and responds to motion along the instrument and elsewhere in the field. This

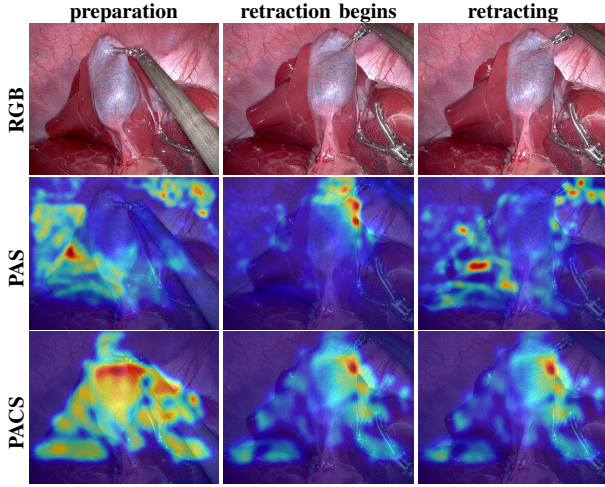


Fig. 2. Pulling-action affordance over time (frames 91, 156, 178). Top: RGB. Middle: kinematic baseline (PAS). Bottom: physics-aware compliance score (PACS). Red = higher affordance, blue = lower. PACS sharpens onto the grasp point and remains stable, while PAS varies more from frame to frame.

matches the intuition that during a pull the surgeon aligns tool and tissue so that the action stays within compliant directions. This is a qualitative observation.

C. Failure Modes and Limits

- (i) The grasped tissue itself can lack affordance in 3D: reconstruction there is noisy under tool occlusion, and furthest-point sampling places few RGP near the contact.
- (ii) Background regions move with the heartbeat, producing non-actionable responses that temporal filtering only partly removes.

D. Does K_i Encode Real Tissue Mechanics?

If the stiffness metric captures real mechanics, observed tissue motion should preferentially follow the compliant direction. For each RGP and frame we compute the absolute cosine between $e_{i,t}^{\min}$ and the normalized next-step displacement $u_{i,t}$ (frames with negligible motion excluded), and report the per-frame median over RGPs. Random directions in $\mathbb{R}^3$ give 0.5 in expectation. As baseline we predict the next direction by the previous one (velocity persistence).

Fig. 3 shows that the baseline hovers around the random level, whereas the XPBD-derived direction reaches roughly 0.55 with 20 solver iterations and roughly 0.75 with 50; removing segmentation-aware filtering at 50 iterations drops it to roughly 0.65 (values read from the plot). More iterations refine the deformation gradient and stabilize K_i , and respecting organ boundaries matters. Velocity persistence is a weak baseline on jittery tracked points, so this supports the physical prior but does not rank it against stronger predictors.

V. DISCUSSION: TOWARD TWINS AND NEXT STEPS

A twin-lite layer. ARG-VU uses the same constraint family as position-based tissue simulators [1], [2], [14], [21], so its K_i estimates could inform stiffness priors for a full simulator. Because $A_{i,t}$ needs only current geometry and

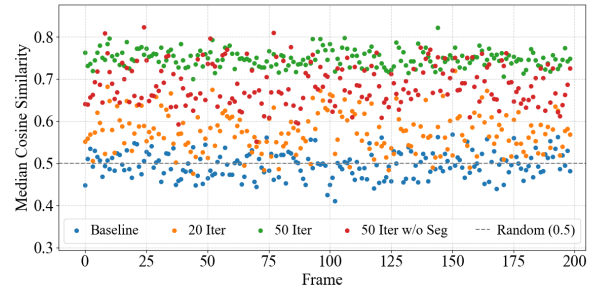


Fig. 3. Per-frame median $|\cos|$ between the predicted compliant direction and observed next-step motion. XPBD-derived directions (20/50 iterations, and 50 without segmentation filtering) versus velocity persistence; dashed line: random level (0.5).

a candidate tool motion, it could also serve as a physics-consistency check on actions proposed by planners, learned policies, or generative and world models. Neither is tested yet. A further step is to turn tracked points into surface meshes that seed full simulation; the mesh quality obtainable from surgical clips is an open question.

Task-conditioned affordance. “High compliance is high affordance” holds for retraction but not for puncture or precise dissection, where surgeons exploit stiffness or tension. The physics backbone supports this: a task-conditioned score could favor compliant directions for retraction, stiffer or higher-tension regions for dissection, and alignment with the constraint normal for injection.

Next steps. (1) Validate on more organs, tools and tasks against expert-annotated affordance. (2) Run in real time and in the loop with a twin.

VI. CONCLUSION

ARG-VU links tracked tissue geometry to action-level reasoning through local XPBD constraint models, scoring whether a tool motion is mechanically admissible without tissue parameters or meshes. Early results support the stiffness prior and show a more stable focus at the grasp than a kinematic baseline.

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